Brainstorming & Idea Prioritization

Date	30 June 2026
Team ID	
Project Name	Rising Waters - Flood Hazard Forecasting System
Maximum Marks	4 Marks

1. Idea Generation & Feature Matrix

Feature	Hydrological Significance	Impact Value	Technical Effort
Cloud Cover (%)	Indicates storm cloud volume & visibility	High	Low
June-Sept Rainfall	Monsoon seasonal total (Key indicator)	Critical	Low (Threshold Rule)
Annual Rainfall	Controls regional baseline water table limits	High	Low
Jan-Feb / Mar-May	Winter & spring precipitation baselines	Medium	Medium

2. Strategic Prioritization Details

We prioritized meteorological features using an Impact vs. Effort matrix. Standard models (like XGBoost or Random Forest) require large system libraries (pandas, scikit-learn, xgboost) that increase bundle zip size on Vercel serverless. We compiled standard StandardScaler parameters and a pure Python Decision Tree threshold (Jun-Sep Rainfall > 2425.64 mm) directly. This provides a 96.55% accuracy model with zero third-party dependencies, running in under 1ms.